

Retail Sales Analytics & Forecasting Report

Executive Business Report

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Tools: Python | PostgreSQL | Power BI | Machine Learning (ARIMA)

Executive Summary

This project delivers an end-to-end retail analytics solution using the Superstore dataset. The objective was to analyze historical sales performance, identify profitability drivers, evaluate regional and product-level performance, and forecast future demand. The project combined data cleaning, exploratory data analysis, SQL-based business analysis, Power BI dashboarding, and machine learning forecasting to support executive decision-making.

Business Objectives

- Evaluate overall business performance using sales, profit, and margin KPIs.
- Identify top-performing and underperforming products.
- Analyze customer, category, regional, and state-level performance.
- Generate actionable recommendations to improve profitability.
- Forecast future sales trends to support planning and budgeting.

Dataset Overview

Dataset: Superstore Retail Dataset

Records: 9,994

Columns: 21

Period: 2014–2017

Data Preparation

Data quality checks confirmed no missing values and no duplicate records. Date fields were converted into datetime format and column names were standardized for SQL and analytics workflows.

Key Performance Indicators

Total Sales: \$2.30M

Total Profit: \$286.40K

Profit Margin: 12.47%

Total Orders: 5,009

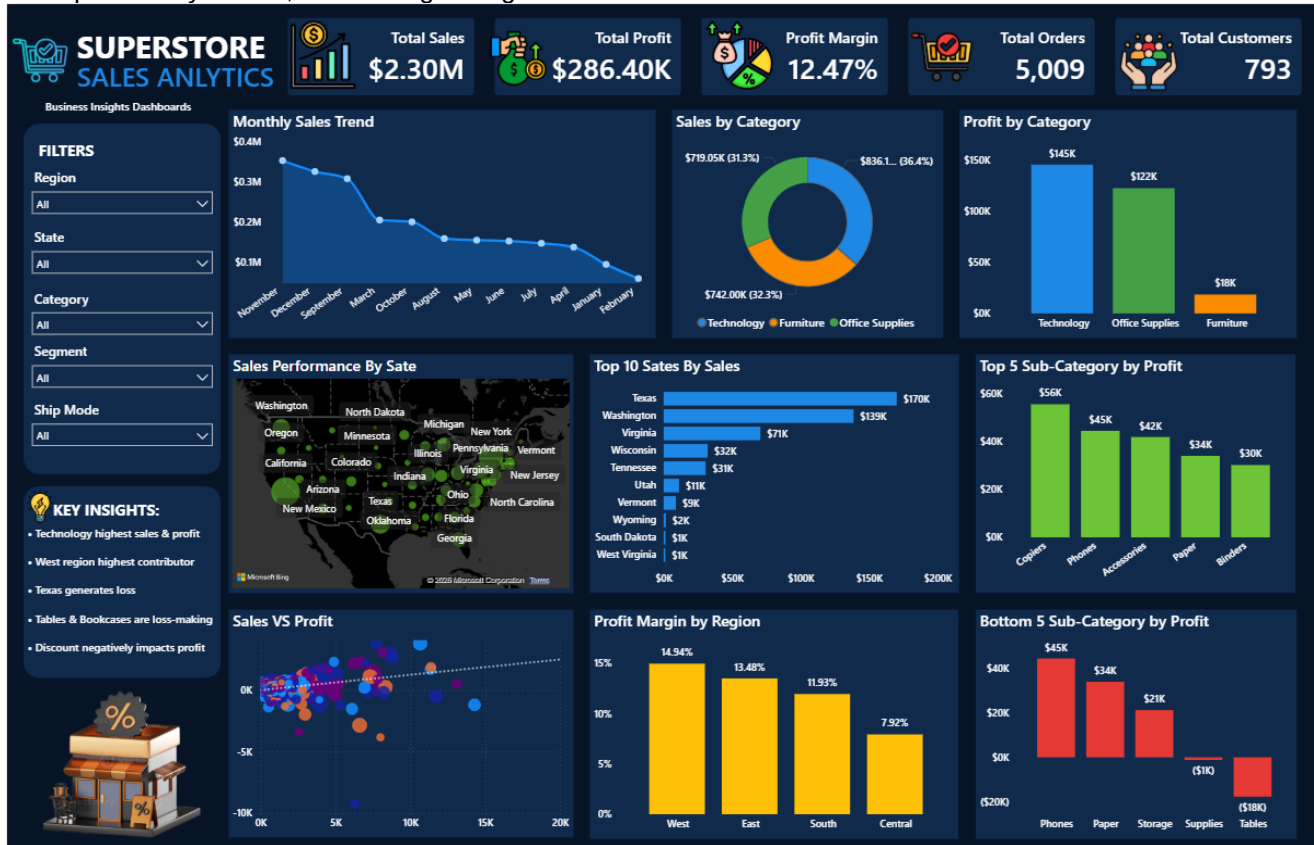
Total Customers: 793

Business Insights

- Technology generated the highest sales and profit among all categories.
- West region was the strongest contributor to revenue and profitability.
- Texas generated high sales volume but produced negative profit.
- Tables and Bookcases were identified as the major loss-making sub-categories.
- Discounts showed a negative relationship with profitability.

Executive Dashboard

A Power BI executive dashboard was developed to provide a consolidated view of business performance. The dashboard includes KPI monitoring, category analysis, geographic analysis, profitability review, and strategic insights.



SQL Analytics & Findings

Key SQL techniques used in the project included Aggregations, CTEs, CASE Statements, Window Functions, Ranking Functions, Growth Analysis, and Business KPI calculations.

Major Findings

Top Profitable Sub-Categories:

- Copiers
- Phones
- Accessories

Loss-Making Sub-Categories:

- Tables
- Bookcases

Customer Segment Analysis:

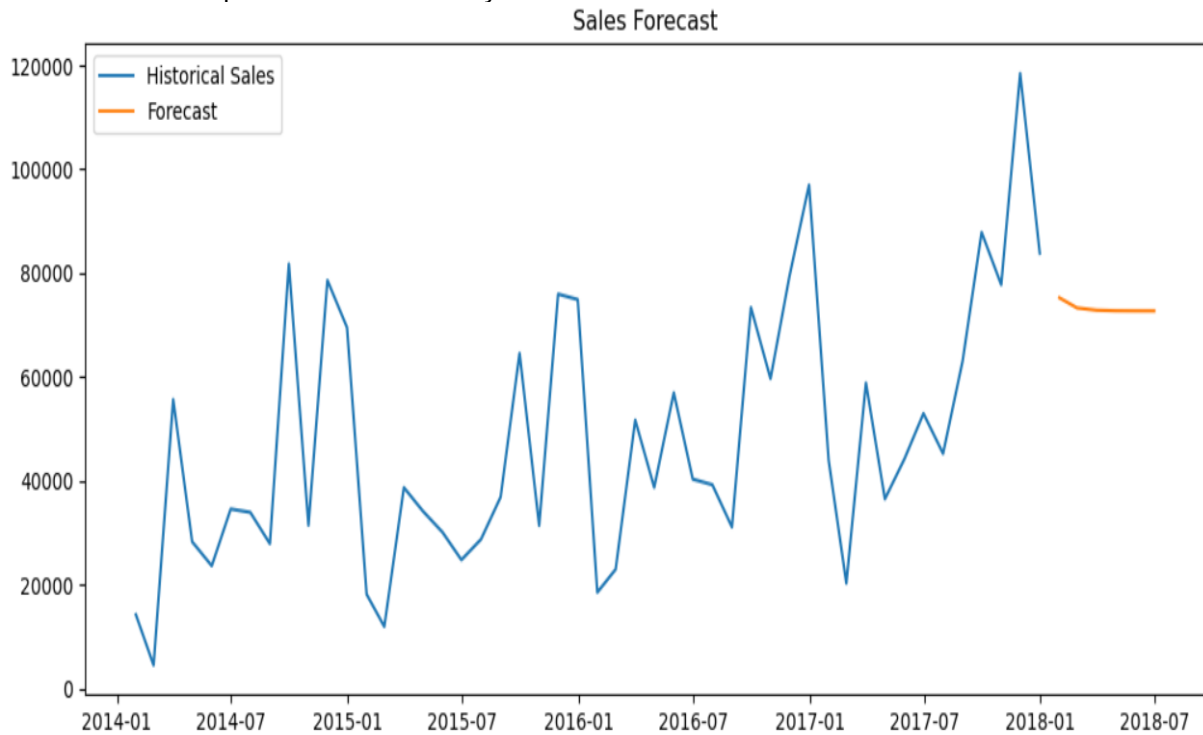
- Consumer segment contributed the highest revenue.
- Standard Class shipping generated the largest share of sales.

Machine Learning Forecasting

A Linear Regression model was initially evaluated but failed to capture the non-linear and seasonal behavior of retail sales data ($R^2 = -1.63$). To address this limitation, an ARIMA(1,1,1) time-series forecasting model was implemented.

Model Diagnostics:

- AIC: 1085.79
- Residual autocorrelation not significant.
- Residual variance remained stable.
- Model assumptions were reasonably satisfied.



Forecasted Monthly Sales:

Jan 2018: \$75,259

Feb 2018: \$73,307

Mar 2018: \$72,862

Apr 2018: \$72,761

May 2018: \$72,738

Jun 2018: \$72,732

Strategic Recommendations

1. Increase investment in Technology products due to strong revenue and profit performance.
2. Reassess discounting strategies to protect profit margins.
3. Investigate operational and pricing issues in Texas and other low-profit regions.
4. Optimize inventory allocation using forecasted demand levels.
5. Reduce exposure to loss-making products such as Tables and Bookcases.
6. Focus marketing spend on high-performing customer segments and regions.

Executive Conclusion

This project successfully transformed raw transactional data into actionable business intelligence. The analysis identified key growth drivers, profitability challenges, and future sales expectations. Technology emerged as the primary growth engine, while several products and regions require profitability improvement initiatives. Forecasting results suggest that monthly sales are likely to remain stable in the \$72K–75K range in the near term, supporting informed planning and resource allocation decisions.